



2019

ممثل

قسم هندسة الطاقة الكهربائية

مصطفى القرشي

اسئلة سنوات سيركت ٢



طلاب البور والاتصالات



spark team



spark-fet.com





Problem 1 [4 points]

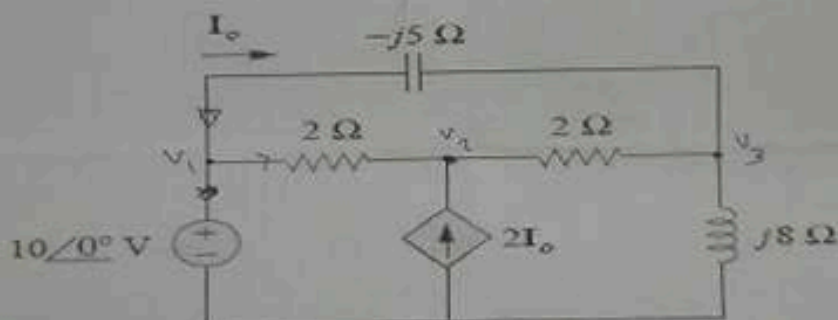
For the following pair of sinusoids, answer the following:

$V_1(t) = 4 \sin(377t + 10^\circ)$ and $V_2(t) = -4 \cos(377t + 10^\circ)$

- 1- V_2 leads V_1 by angle of 27°
- 2- The voltage (v_1) value at $t = 5 \mu s$ is: $-3.923 V$ $16.97 V$
- 3- The frequency in Hertz is: $60 / 2\pi = 9.549$ $16.97 V$
- 4- The period in second is: $1 / 9.549 = 0.1047$

Problem 2 [8 points]

For the circuit shown in figure below :



a) Write the nodal equations for the circuit?

$V_1 - 10 \angle 0^\circ = 0$... ①

$\frac{V_1 - V_2}{2} + 2I_0 = \frac{V_1 - V_3}{-j5}$

$\frac{V_2 - V_3}{2} + \frac{V_3 - V_1}{j8} = \frac{V_2}{-j5}$

b) Find the values of :

$V_1 = 10 V$

$V_2 =$ _____

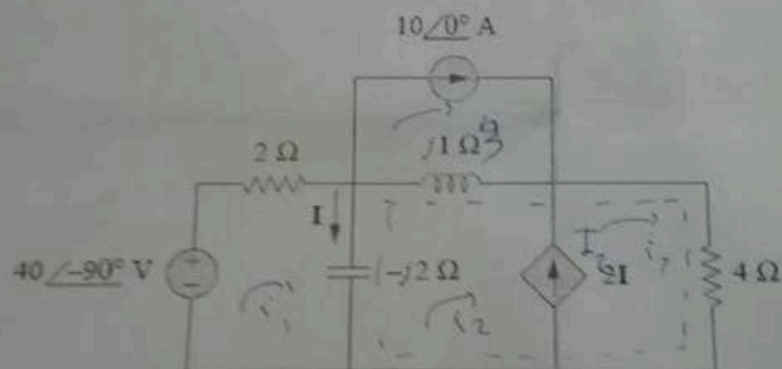
$V_3 =$ _____

6/8

$I_0 =$ _____

Problem 3 [8points]

For the circuit shown below.



a) Write the mesh equations for the circuit?

[illegible]

Pro 1 :- $V_1(t) = 4 \sin(377t + 10^\circ) \text{ V}$
 $V_2(t) = -4 \cos(377t + 10^\circ) \text{ V}$

1

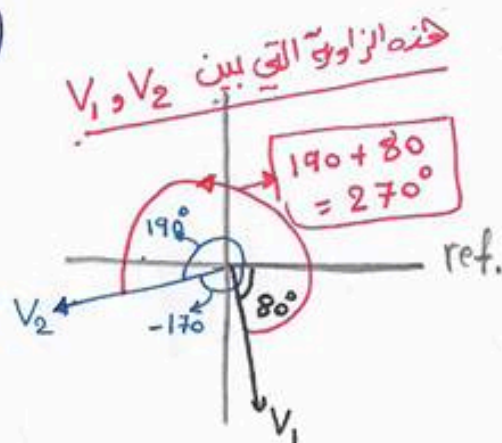
$\rightarrow V_1(t) = 4 \cos(377t + 10^\circ - 90^\circ)$

$\rightarrow V_2(t) = 4 \cos(377t + 10^\circ \pm 180^\circ)$

$\rightarrow V_1(t) = 4 \cos(377t - 80^\circ)$

$\rightarrow V_2(t) = 4 \cos(377t - 170^\circ)$

$\rightarrow$ or $V_2(t) = 4 \cos(377t + 190^\circ)$



① V_2 lead V_1 by 270° ✓

② The Voltage (V_1) Value at $t = 5 \text{ ms}$ is: 3.53 Volt ✓

$V_1(t) = 4 \cos(377t - 80^\circ)$

$\rightarrow V_1(5 \text{ ms}) = 4 \cos(377 * 5 * 10^{-3} \text{ ms} - 80^\circ)$
 $= 4 \cos(1.885 - 80^\circ)$ حولها الى degree

$= 4 \cos(108 - 80) = 3.53$

$\text{rad} * \frac{180}{\pi} = \text{deg}$

$\Rightarrow 1.885 * \frac{180}{\pi} = 108^\circ$

③ The frequency in Hertz is: 60 Hz

$f = \frac{1}{T} = \frac{2\pi}{2\pi} = \frac{377}{2\pi} = 60 \text{ Hz}$

④ The period in second is: 0.01667 sec ✓

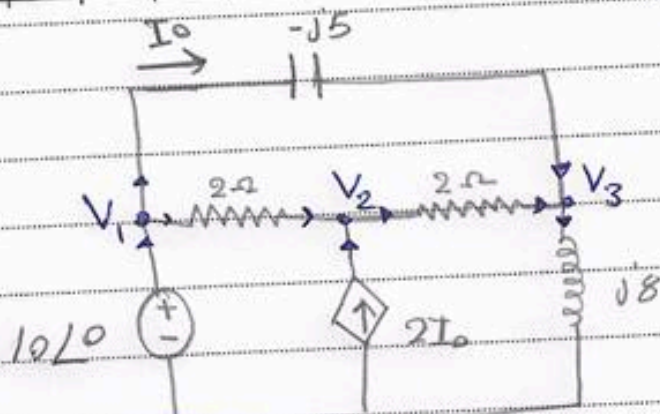
$T = \frac{1}{f} = \frac{1}{60} = 0.01667$

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 #أبو-الشيخ-واقظ

(2)

Subject

Prob. 2



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 إن شاء الله تعالى

a

$$V_1 = 10 \angle 0 \quad (1)$$

$$\rightarrow \frac{V_1 - V_2}{2} + 2I_0 = \frac{V_2 - V_3}{2} \rightarrow \frac{10 \angle 0 - V_2}{2} + 2I_0 = \frac{V_2 - V_3}{2} \quad (2)$$

$$\rightarrow \frac{V_2 - V_3}{2} + \frac{V_1 - V_3}{-j5} = \frac{V_3}{j8} \rightarrow \frac{V_2 - V_3}{2} + \frac{10 \angle 0 - V_3}{-j5} = \frac{V_3}{j8} \quad (3)$$

b

$$V_1 = 10 \angle 0$$

$$V_2 \text{ and } V_3$$

$$= j4 - j0.4V_3$$

$$\text{From eq 2: } 2I_0 \Rightarrow I_0 = \frac{V_1 - V_3}{-j5} \Rightarrow \left(\frac{10 \angle 0 - V_3}{-j5} \right) \times 2$$

$$\Rightarrow \left(\frac{10 \angle 0 - V_2}{2} + j4 - j0.4V_3 = \frac{V_2 - V_3}{2} \right) \times (10) \text{ بالهزب في } (10)$$

$$\rightarrow 50 - 5V_2 + j40 - j4V_3 = 5V_2 - 5V_3$$

$$\rightarrow (-10V_2 + (5 - j4)V_3 = -50 - j40) \quad (1)$$

Subject

From eq (3) $\left(\frac{V_2 - V_3}{2} + \frac{10 \angle 0^\circ - V_3}{-j5} - \frac{V_3}{j8} \right) * (j40)$ بالضرب في (40j)

$$\rightarrow j20V_2 - j20V_3 + 80 + 8V_3 = 5V_3$$

$$\rightarrow j20V_2 + (8 - j20)V_3 = 80 \quad (2)$$

$$\begin{bmatrix} -10 & 5-j4 \\ j20 & 3-j20 \end{bmatrix} \begin{bmatrix} V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -50-j40 \\ 80 \end{bmatrix}$$

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#أنت الحل - وانا فقط

$$\Delta = (-10 * (3-j20)) - (j20 * (5-j4)) = (-110 + j100)$$

$$\Delta_2 = \begin{vmatrix} -50-j40 & 5-j4 \\ 80 & 3-j20 \end{vmatrix} = [(-50-j40) * (3-j20)] - [(5-j4) * 80]$$

$$\Delta_2 = (-1350 + j1200)$$

$$\Delta_3 = \begin{vmatrix} -10 & -50-j40 \\ j20 & 80 \end{vmatrix} = (-10 * 80) - ((-50-j40) * j20)$$

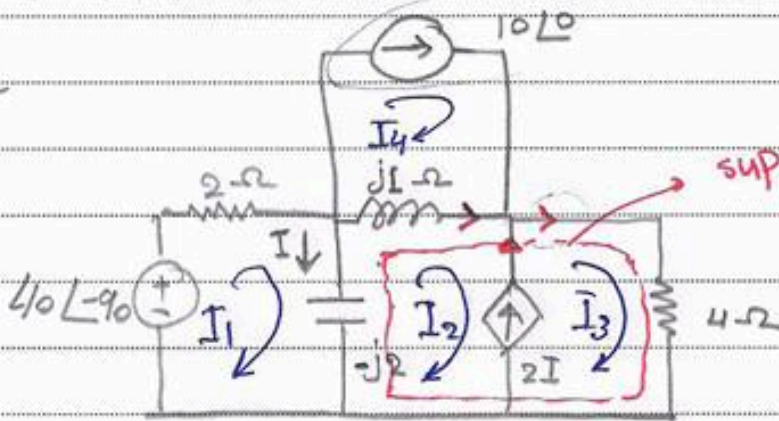
$$\Delta_3 = -1600 + j1000$$

$$\therefore V_2 = \frac{\Delta_2}{\Delta} = \frac{(-1350 + j1200)}{(-110 + j100)} = 12.15 + j0.136 \text{ Volt}$$

$$= 12.15 \angle 0.64^\circ \text{ Volt}$$

$$V_3 = \frac{\Delta_3}{\Delta} = \frac{-1600 + j1000}{-110 + j100} = 12.49 + j2.26 \text{ Volt}$$

$$= 12.7 \angle 10.25^\circ \text{ Volt}$$

Prob 3

(a) super mesh eq: $I_2 + 2I = I_3$; $I = I_1 - I_2$

① $-I_2 + 2I_1 - I_3 = 0$

Mesh 1 → $j40 + (2 + j2)I_1 + j2I_2 = 0$ بالقوة على 2

② $(1 - j)I_1 + jI_2 = -j20$

Mesh 4 → $I_4 = 10 \angle 0^\circ \text{ A}$ ③

Mesh 2, 3 → $-jI_2 + j2I_1 - jI_4 + 4I_3 = 0$

$I_4 = 10 \angle 0^\circ$

$I_3 = I_2 + 2I_1$ ④

$4I_3 = 4I_2 + 8I_1$

$-jI_2 + j2I_1 - j10 - 4I_2 + 8I_1 = 0$

④' $(8 + j2)I_1 + (-4 - j)I_2 = j10$

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#أبدي - أروع - وأفضل

Subject _____

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عالم الفيزياء

(5)

(b) $I_1, I_2, I_3, I_4 = ???$

$I_4 = 10 \angle 0^\circ$ ~~(-1)~~

From eq. 2 + 4

$$\begin{bmatrix} 1-j & j \\ 8+j2 & (-4-j) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -j20 \\ j10 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 1-j & j \\ 8+j2 & (-4-j) \end{vmatrix} = [(1-j) \times (-4-j)] - [(8+j2) \times j]$$

$$\Delta = -3 - j5$$

$$\Delta_1 = \begin{vmatrix} -j20 & j \\ j10 & (-4-j) \end{vmatrix} = [(-j20) \times (-4-j)] - (j10 \times j)$$

$$\Delta_1 = -10 + j80$$

$$\Delta_2 = \begin{vmatrix} 1-j & -j20 \\ 8+j2 & j10 \end{vmatrix} = [(1-j) \times j10] - [(-j20) \times (8+j2)]$$

$$\Delta_2 = -30 + j170$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{-10 + j80}{-3 - j5} = -10.88 - j8.53 \text{ A}$$

$$I_1 = 13.83 \angle -141.9^\circ \text{ A}$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{-30 + j170}{-3 - j5} = 29.6 \angle -139.02^\circ \text{ A}$$

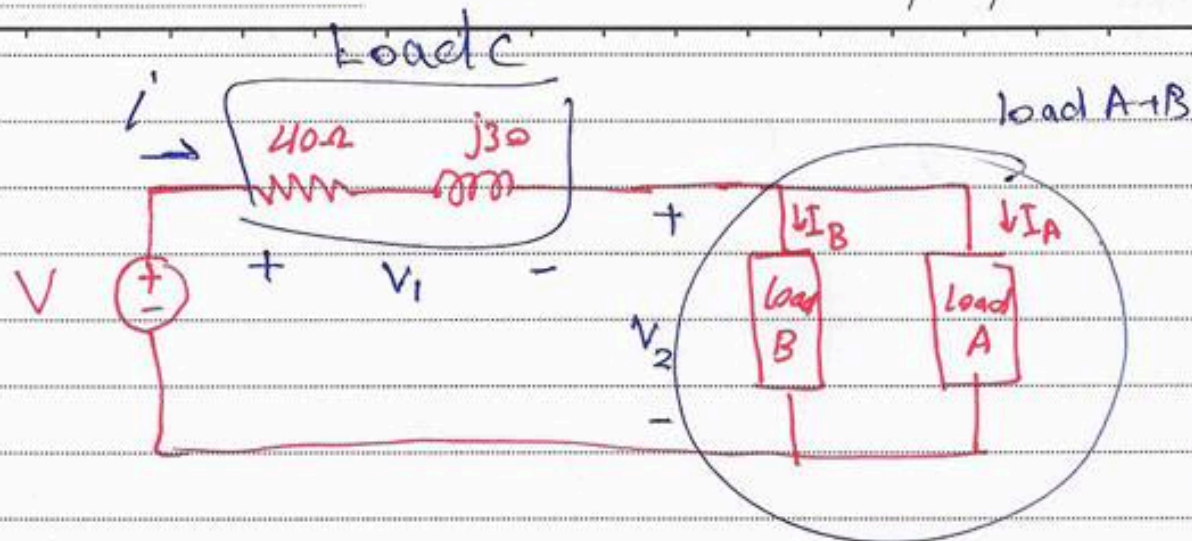
$$I_3 = -I_2 + 2I_1 = 2.415 \angle 76.11^\circ \text{ A}$$

$$I_4 = 10 \angle 0^\circ$$

(6)

Subject

Q3



Load	S (VA)	P (W)	Q (VAR)	ϕ (deg)	PF (lead/lag)	I (A)	V_{rms}
A	250	200	$-j150$	-36.869°	0.8 lead	1	250
B	300	Zero	$j300$	90°	0 lag	1.2	250
A+B	250	200	$j150$	36.87°	0.8 lag	1	250
C	50	40	$j30$	36.869°	0.8 lag	1	50
A+B+C	300	240	$j180$	36.869°	0.8 lag	1	300

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Subject

③ $S = P + jQ = 200 + j150 = \sqrt{200^2 + 150^2} = 250 \text{ VA}$

Subject

/ /

$$\textcircled{4} \quad \phi = \tan^{-1}\left(\frac{Q}{P}\right) = \tan^{-1}\left(\frac{150}{200}\right) = +36.87^\circ$$

$$\textcircled{5} \quad \text{P.f} = \cos(\phi) = \cos(36.87) = 0.8 \text{ lag}$$

$$\textcircled{6} \quad \boxed{V_{r.m.s} = 250 \text{ V}} \rightarrow \text{Parallel كوازي}$$

$$\textcircled{7} \quad I_{r.m.s} = \frac{S}{V_{r.m.s}} = \frac{250}{250} = 1 \text{ A}$$

load c $I_{r.m.s} = 1$, $R = 40 \Omega$, $X_L = j30 \Omega$

$$\textcircled{1} \quad P = I_{r.m.s}^2 \times R = 1^2 \times 40 = 40 \text{ W}$$

$$\textcircled{2} \quad Q = I_{r.m.s}^2 \times X_L = 1^2 \times j30 = j30$$

$$\textcircled{3} \quad S = P + jQ = 40 + j30 = \sqrt{40^2 + 30^2} = 50$$

$$\textcircled{4} \quad \phi = \tan^{-1}\left(\frac{Q}{P}\right) = \tan^{-1}\left(\frac{30}{40}\right) = +36.869^\circ$$

$$\textcircled{5} \quad \text{P.f} = \cos(\phi) = \cos(36.869) = 0.8 \text{ lag}$$

$$\textcircled{6} \quad S = VI \Rightarrow V = \frac{S}{I} = \frac{50}{1} = 50 \text{ V}$$

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Load A+B+C

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$$① P_T = 200 + 0 + 40 = 240 \text{ W}$$

$$② Q_T = -j150 + j300 + j30 = j180$$

$$③ S_T = \sqrt{240^2 + 180^2} = 300 \text{ VA}$$

$$④ \theta = \tan^{-1}\left(\frac{Q}{P}\right) = \tan^{-1}\left(\frac{180}{240}\right) = +36.86^\circ$$

$$⑤ P.F. = \cos(\theta) = \cos(36.86) = 0.8$$

Lag

$$⑥ V_{r.m.s} = \frac{V_{A+B} + V_C}{T} = 250 + 50 = 300 \text{ V}$$

توالي
التيار على التوالي لذلك يرجع

$$⑦ S = V_{r.m.s} I_{r.m.s} \Rightarrow I_{r.m.s} = \frac{S}{V_{r.m.s}}$$

$$\Rightarrow I_{r.m.s} = \frac{300}{300} = 1 \text{ A}$$

Problem 1 [8 points]

For the following pair of sinusoids, answer the following:

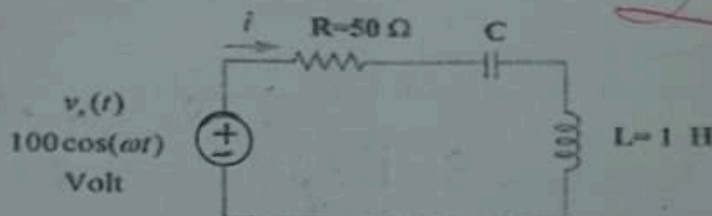
$V_1(t) = 4 \sin(377t + 10^\circ)$ and $V_2(t) = -4 \cos(377t + 10^\circ)$

- 1- V_2 leads V_1 by angle of 110°
- 2- The voltage (v_1) value at $t = 5 \mu s$ is: 3.99
- 3- The frequency in Hertz is: $60,000$
- 4- The period in second is: 1.66×10^{-3}

Consider the circuit below which is operating with a sinusoidal source.

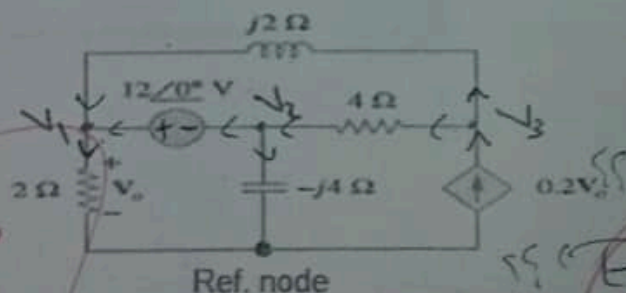
5. The value of the capacitance C , if $i(t) = \sqrt{2} \cos(\omega t - 45^\circ)$ A and $\omega = 100$ rad/sec. -50

6. The value of ω , if $i(t) = \sqrt{2} \cos(\omega t + 45^\circ)$ A and $C = (5/3)$ mF. 25



Problem 2 [8 points]

For the circuit shown in figure below :



a) Write the nodal equations for the circuit?

$$12 = V_1 - V_2$$

$$(2 + j2)V_1 + (-j4)V_2 + 0.2 = 0$$

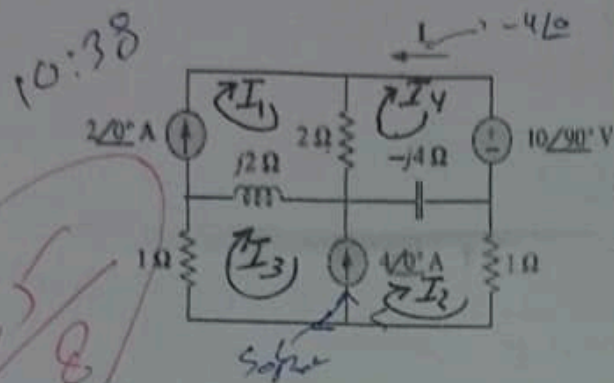
$$(-j4)V_1 + (4 + j2)V_2 + 0.2 = 0$$

b) Find the values of:

$$V_1 = 4.615 \quad V_2 = -2.36 \quad V_3 = -2.58 + j0$$

Problem 3 [8 points]

For the circuit shown in figure below :



a) Write the mesh equations for the circuit?

$$\begin{aligned} I_2 &= 4/0 + I_3 \\ (2+j2)I_1 + 2I_4 - 2jI_3 &= 0 \\ -2I_1 - 4jI_2 + (2-j4)I_4 &= 10/90 \\ -2jI_1 + (1-j4)I_2 + (1+j2)I_3 - 4jI_4 &= 0 \end{aligned}$$

b) Find the values of:

$$\begin{aligned} I_1 &= 2/0 \text{ A} & I_2 &= 6-6j \text{ A} & I_3 &= 2-6j \text{ A} \\ I_4 &= -4/0 = -4/0 \text{ A} \end{aligned}$$

Problem 4 [10 points]

For the circuit shown in figure below

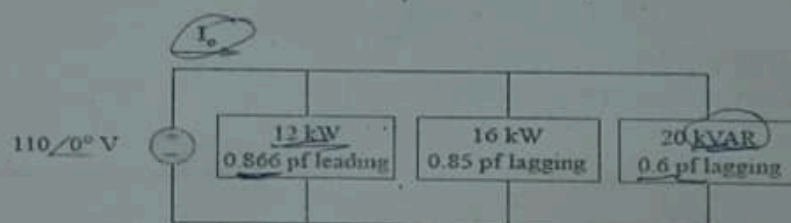
The total active power is 23.83 W

The total reactive power is 0 VAR

The total complex power is $S = P + jQ = 23.8$

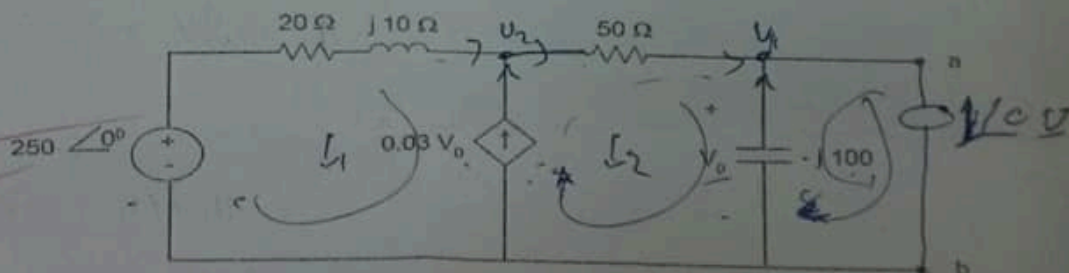
The power factor at the source is 1

The value of I_0 is 1.4334 A



Problem 5 [6 points]

Find the Thevenin's equivalent between a and b



Q1 A

$$V_1(t) = 4 \sin(377t + 10^\circ)$$

$$V_2(t) = -4 \cos(377t + 10^\circ)$$

Ahmed Al-Khazri

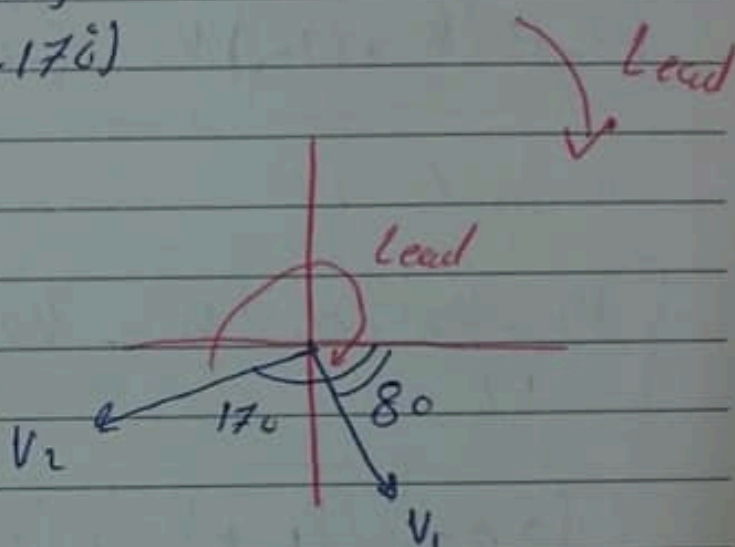
① V_2 lead V_1 by 270°

$$V_1(t) = 4 \cos(377t + 80^\circ)$$

$$V_2(t) = 4 \cos(377t - 170^\circ)$$

$$170 - 80 = 90$$

$$360 - 90 = \boxed{270}$$



② $V_1(\text{rms}) = 4 \sin(377(\text{ms}) + 10^\circ)$

محول من رار الى دوسر

$$\text{deg} = \frac{180}{\pi} \text{ rad}$$

$$\text{deg} = \frac{180}{\pi} \times 377 \times \text{ms} = 0.108$$

$$\boxed{V_1 = 0.7 \text{ Volt}}$$

③ $f = \frac{\omega}{2\pi} = \frac{377}{2\pi} = \boxed{60 \text{ Hz}}$

④ $T = \frac{1}{f} = \frac{1}{60} = \boxed{0.0166 \text{ s}}$

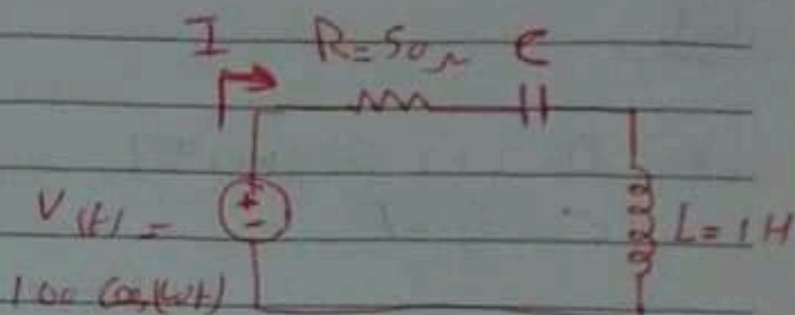
Q1

B

(a) $i(t) = \sqrt{2} \cos(\omega t - 45^\circ)$

$\omega = 100 \text{ rad/sec}$

Find C ??



Sol:- $V(t) = 100 \angle 0^\circ$ $I(t) = \sqrt{2} \angle -45^\circ$

$X_L = j\omega L = j(100)(1) = j100 \rightarrow X_L$

$V(t) = I(t) Z \rightarrow Z = \frac{V(t)}{I(t)} = \frac{100 \angle 0^\circ}{\sqrt{2} \angle -45^\circ}$

$\rightarrow Z = 50 + j50$

$Z = R + X_L + X_C$

$50 + j50 = 50 + j100 + X_C \rightarrow X_C = -j50$

$X_C = \frac{1}{j\omega C} \rightarrow -j50 = \frac{1}{j(100)C}$

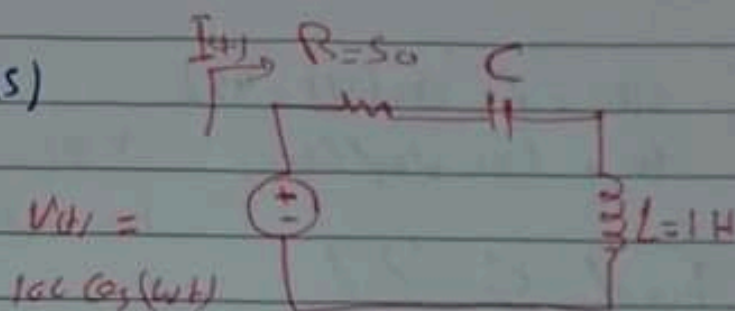
$\rightarrow C = 2 \times 10^{-4} \text{ F}$

Ahmad Al-Khawaja

Q1

(b) $I(t) = \sqrt{2} \cos(\omega t + 45^\circ)$
 $C = \frac{5}{3} \mu F$

Find ω ?



Sol:-

$$X_L = j\omega L = j\omega$$

$$X_C = \frac{1}{j\omega C} = \frac{-j600}{\omega}$$

By Ohm's law $V(t) = I(t) Z \rightarrow Z = \frac{V(t)}{I(t)} = \frac{100 \angle 0^\circ}{\sqrt{2} \angle 45^\circ}$

$$Z = 50 - j50$$

$$Z = R + X_L + X_C \rightarrow 50 - j50 = 50 + j\omega - \frac{j600}{\omega}$$

$$\rightarrow j\omega - \frac{j600}{\omega} + j50 = 0$$

$$\rightarrow \left[\omega - \frac{600}{\omega} + 50 = 0 \right] \omega$$

Ahmed Al-Khazragi

$$\rightarrow \omega^2 + 50\omega - 600 = 0$$

$$(\omega + 60)(\omega - 10) = 0$$

~~or~~ $\omega = -60$

$$\omega = 10$$

2L is ω

~~X~~

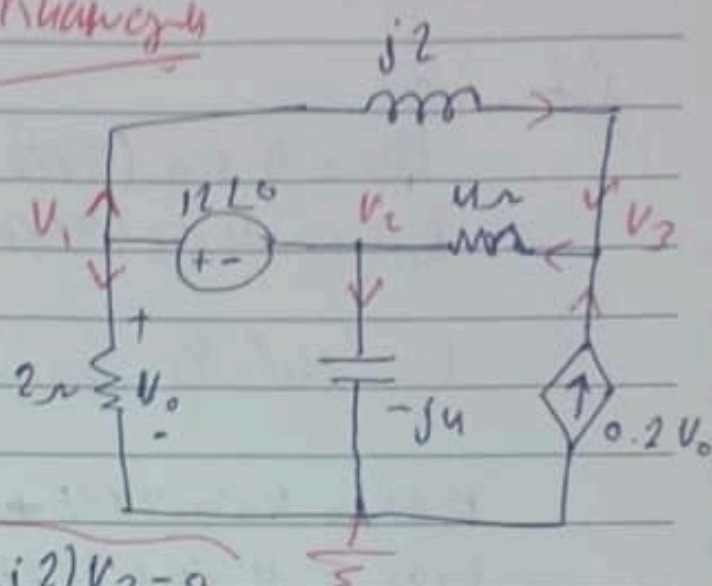
$$\omega = 10 \text{ rad/sec}$$

Q2

Abmuel Al-Khamezi

at node ①+②

$$4 \left[\frac{V_3 - V_2}{4} = \frac{V_1}{2} + \frac{V_2}{-j4} + \frac{V_1 - V_3}{2j} \right]$$



$$(2 - j2)V_1 + (1 + j)V_2 + (-1 + j2)V_3 = 0 \quad \text{--- ①}$$

at node ③

$$V_0 = V_1$$

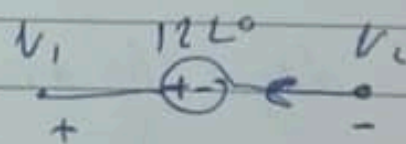
$$\left[\frac{V_1 - V_3}{j2} + 0.2V_1 = \frac{V_3 - V_2}{4} \right] \quad 4$$

$$(0.8 - j2)V_1 + V_2 + (-1 + j2)V_3 = 0 \quad \text{--- ②}$$

at super node

$$-V_2 - 12 + V_1 = 0$$

$$V_1 - V_2 = 12 \quad \text{--- ③}$$



$$\begin{bmatrix} 2 - j2 & 1 + j & -1 + j2 \\ 0.8 - j2 & 1 & -1 + j2 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$$

$$V_1 = 7.68 \angle 50.19^\circ$$

$$V_2 = 9.21 \angle 146.19^\circ$$

$$V_3 = 3.88 \angle 68.62^\circ$$

Ahmed Al-Khazragi

Q3

at mesh (1)

$$I_1 = 2 \angle 0^\circ \text{ A}$$

at mesh (4)

$$2(I_4 - I_1) + 10 \angle 90^\circ + (-j4)(I_4 - I_2) = 0$$

كوكب فية (I_1) في الكبار

$$\rightarrow j4 I_2 + (2 - j4) I_4 = 4 - j10 \quad \text{--- (1)}$$

at mesh (2) + (3)

$$I_3 + (j2)(I_3 - I_1) + (-j4)(I_2 - I_4) + I_2 = 0$$

كوكب فية (I_1) في الكبار

$$(1 - j4) I_2 + (1 + j2) I_3 + (j4) I_4 = j4 \quad \text{--- (2)}$$

at super mesh

$$I_2 = 4 \angle 0^\circ + I_3 \rightarrow I_2 - I_3 = 4 \angle 0^\circ \quad \text{--- (3)}$$

$$\begin{bmatrix} j4 & -j4 & 2 + j4 \\ 1 - j4 & 1 + j2 & j4 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 4 - j10 \\ j4 \\ 4 \angle 0^\circ \end{bmatrix}$$

$$I_1 = 2 \angle 0^\circ \text{ A}$$

$$I_2 = 1.05 \angle 18.4^\circ \text{ A}$$

$$I_3 = 3.018 \angle 173.65^\circ \text{ A}$$

$$I_4 = 3.34 \angle -5.71^\circ \text{ A}$$

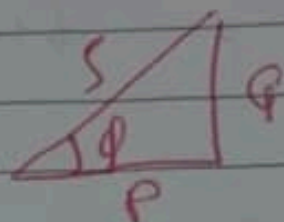
Q4

Load 1 $P_1 = 12 \text{ kW}$ 0.866 Lead

$$\phi_1 = \cos^{-1}(0.866) = 30^\circ$$

$$Q_1 = P_1 \tan \phi_1 = 6.928 \text{ KVAR}$$

$$\tilde{S}_1 = P_1 - jQ_1 = 12 \text{ K} - j6.928 \text{ K VA}$$



Load 2 $P_2 = 16 \text{ kW}$ 0.85 Lag

$$\phi_2 = \cos^{-1}(0.85) = 31.78^\circ$$

$$Q_2 = P_2 \tan \phi_2 = 9.915 \text{ KVAR}$$

$$\tilde{S}_2 = P_2 + jQ_2 = 16 \text{ K} + j9.915 \text{ K VA}$$

Load 3 $Q_3 = 20 \text{ KVAR}$ 0.6 Lag

$$\phi_3 = \cos^{-1}(0.6) = 57.13^\circ$$

$$P_3 = \frac{Q_3}{\tan \phi_3} = 15 \text{ kW}$$

$$\tilde{S}_3 = P_3 + jQ_3 = 15 \text{ K} + j20 \text{ K VA}$$

$$\textcircled{3} \quad \tilde{S}_{\text{tot}} = \tilde{S}_1 + \tilde{S}_2 + \tilde{S}_3 = 47 \text{ K} + j22.987 \text{ K VA}$$

$$\textcircled{1} \quad P_{\text{tot}} = 47 \text{ kW}$$

$$\textcircled{2} \quad Q_{\text{tot}} = 22.987 \text{ KVAR}$$

$$S_{\text{tot}} = 48.758 \angle 28.128$$

$$\textcircled{4} \quad \text{PF} = \cos(28.128) = 0.88$$

Q4

⑤

Lo??

$$\tilde{S} = \tilde{V}_{rms} \tilde{I}_{rms}^* = \frac{1}{2} \tilde{V} \tilde{I}^*$$

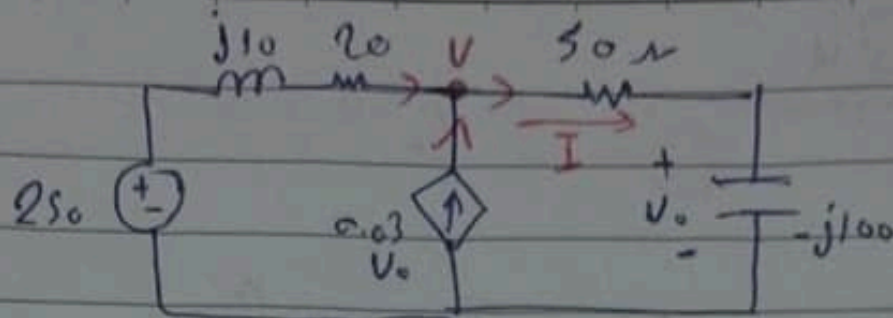
$$\tilde{I} = \left(\frac{2\tilde{S}}{\tilde{V}} \right)^* = (886.52 \angle 28.13)^*$$

$$\tilde{I} = 886.52 \angle -28.13 \text{ A}$$

Amad Al-Khawaja

Q5

A



at node V

$$V_o = V_{th}$$

$$V_o = I(-j100)$$

$$V_o = \frac{V}{50-j100} \times (-j100)$$

$$\frac{250 \angle 0^\circ}{20+j10} - V + 0.03 \left(\frac{V}{50-j100} \times (-j100) \right) = \frac{V}{50-j100}$$

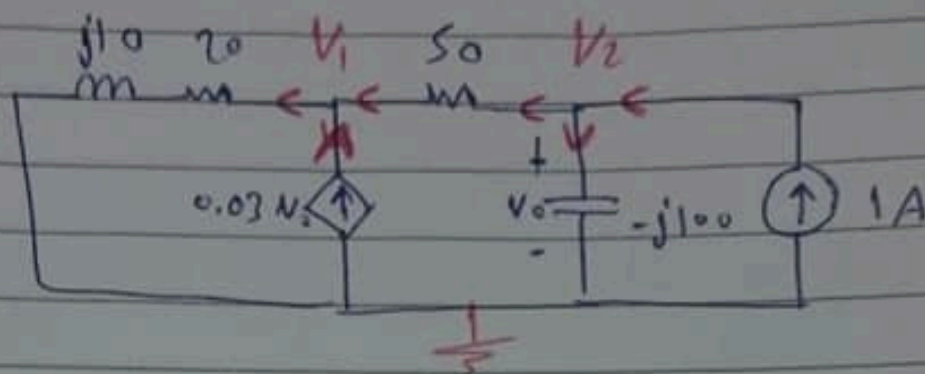
2, leki kharid

$$-0.02 V = -10+j5 \rightarrow V = 500-j250$$

$$V_o = V_{th} = \frac{V}{50-j100} \times (-j100) = 300-j400 \text{ V}$$

Ahmed Al-Khawas

Q5 B



at node ①

$$V_0 = V_2$$

$$\frac{V_2 - V_1}{50} + 0.03V_2 = \frac{V_1}{20 + j10}$$

بقیمه اگه

$$(-0.06 + j0.02)V_1 + (0.05)V_2 = 0 \quad \text{--- ①}$$

at node ②

$$\frac{V_2 - V_1}{50} + \frac{V_2}{-j100} = 1 \rightarrow (-0.02)V_1 + (0.02 + j0.01)V_2 = 1$$

$$\begin{bmatrix} -0.06 + j0.02 & 0.05 \\ -0.02 & 0.02 + j0.01 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$V_1 = 100 - j50$$

$$V_2 = 100 - j100$$

اگر مخرج و ولتاژ را در یکدیگر ضرب کنیم اگه (100-j100) باشد

$$Z_{th} = \frac{V_2}{I_s} = \frac{100 - j100}{1} = 100 - j100$$